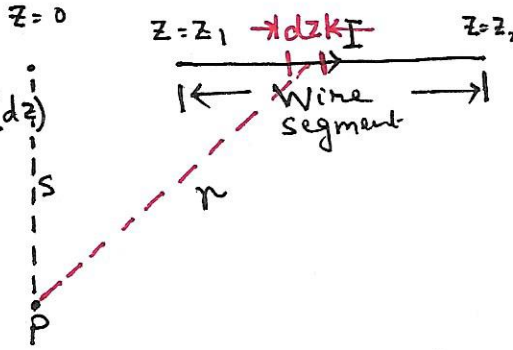
Tutorial 12

1. Let the wire be on z-axis and its length spans from $z = z_1$ to $z = z_2$. Consider an element (dz) of current-carrying wire at a distance z from $z=0$. We would like to calculate vector potential at P due to current element dz



$$\vec{A} = \frac{\mu_0}{4\pi}\int \frac{I\,d\vec{l}'}{r}$$. Here I = constant, can be taken out of integral

$d\vec{l} = dz(\hat{z})$

$$\vec{A} = \frac{\mu_0 I}{4\pi}\int_{z=z_1}^{z_2} \frac{dz}{\sqrt{s^2+z^2}}(\hat{z}) = \frac{\mu_0 I}{4\pi}\ln\left|z+\sqrt{s^2+z^2}\right|_{z_1}^{z_2}(\hat{z})$$

$$= \frac{\mu_0 I}{4\pi}\ln\left(\frac{z_2+\sqrt{s^2+z_2^2}}{z_1+\sqrt{s^2+z_1^2}}\right)\hat{z}$$. We will calculate $\vec{B}$ due to $\vec{A}$ and check whether it matches with $\vec{B}$ due to finite straight wire as derived in the class. Use the relation $\vec{B} = \vec{\nabla}\times\vec{A}$ and use expression of curl in cylindrical co-ordinates (s, φ, z)

$$\vec{\nabla}\times\vec{A} = \left(\frac{1}{s}\frac{\partial A_z}{\partial \varphi} - \frac{\partial A_\varphi}{\partial z}\right)\hat{s} + \left(\frac{\partial A_s}{\partial z} - \frac{\partial A_z}{\partial s}\right)\hat{\varphi} + \frac{1}{s}\left[\frac{\partial}{\partial s}(s A_\varphi) - \frac{\partial A_s}{\partial \varphi}\right]\hat{z}$$

But in this problem, we only have $A_z(s)$

$$\vec{\nabla}\times\vec{A} = \vec{B} = -\frac{\partial A_z}{\partial s}\hat{\varphi}$$

$$\vec{B} = -\frac{\mu_0 I}{4\pi}\left[\frac{z_1+\sqrt{s^2+z_1^2}}{z_2+\sqrt{s^2+z_2^2}} \times \left\{\frac{\left(z_1+\sqrt{s^2+z_1^2}\right)\times\frac{2s}{2\sqrt{s^2+z_2^2}} - \left(z_2+\sqrt{s^2+z_2^2}\right)\frac{s}{\sqrt{s^2+z_1^2}}}{\left(z_1+\sqrt{s^2+z_1^2}\right)^2}\right\}\right]\times\hat{\varphi}$$

$$= -\frac{\mu_0 I}{4\pi}\left[\frac{1}{z_2+\sqrt{s^2+z_2^2}}\left(\frac{s}{\sqrt{s^2+z_2^2}}\right) - \frac{s}{\sqrt{s^2+z_1^2}}\left(\frac{1}{z_1+\sqrt{s^2+z_1^2}}\right)\right]\hat{\varphi}$$

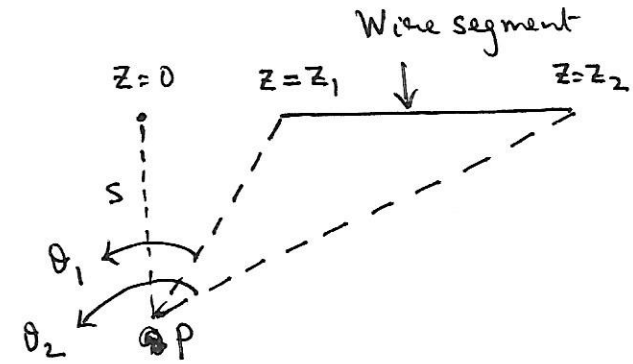
Multiply and divide by $\left(z_2-\sqrt{s^2+z_2^2}\right)$

Multiply and divide by $\left(z_1-\sqrt{s^2+z_1^2}\right)$

$$\vec{B} = -\frac{\mu_0 I}{4\pi}\hat{\varphi}\left[\frac{s}{\sqrt{s^2+z_2^2}}\frac{\left(z_2-\sqrt{s^2+z_2^2}\right)}{\left[\cancel{z_2^2}-(s^2+\cancel{z_2^2})\right]} - \frac{s}{\sqrt{s^2+z_1^2}}\frac{\left(z_1-\sqrt{s^2+z_1^2}\right)}{\left[\cancel{z_1^2}-(s^2+\cancel{z_1^2})\right]}\right]$$

$$= \frac{\mu_0 I s}{4\pi s^2}\left[\frac{1}{\sqrt{s^2+z_2^2}}\left(z_2-\sqrt{s^2+z_2^2}\right) - \frac{1}{\sqrt{s^2+z_1^2}}\left(z_1-\sqrt{s^2+z_1^2}\right)\right]\hat{\varphi}$$

$$\vec{B} = \frac{\mu_0 I}{4\pi s}\left[\underbrace{\frac{z_2}{\sqrt{s^2+z_2^2}}}_{\sin\theta_2} - \underbrace{\frac{z_1}{\sqrt{s^2+z_1^2}}}_{\sin\theta_1}\right]\hat{\varphi}$$



$$\boxed{\vec{B} = \frac{\mu_0 I}{4\pi s}\left(\sin\theta_2 - \sin\theta_1\right)\hat{\varphi}}$$

which tallies with the expression of $\vec{B}$ for a finite length current-carrying wire obtained using Biot-Savart law, as done in class.

2. Given $\vec{A} = k\hat{\varphi}$ (k = constant), we have to find current density $\vec{J}$. Use Ampere's law $\vec{\nabla}\times\vec{B} = \mu_0\vec{J}$

i.e $\vec{J} = \frac{1}{\mu_0}(\vec{\nabla}\times\vec{B}) = \frac{1}{\mu_0}\left[\vec{\nabla}\times(\vec{\nabla}\times\vec{A})\right]$

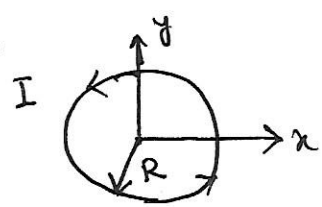
$$\vec{\nabla}\times\vec{A} = \frac{1}{s}\frac{\partial}{\partial s}(sA_\varphi)\hat{z} = \frac{1}{s}\frac{\partial}{\partial s}(sk)\hat{z} = \frac{k}{s}\hat{z}$$

[see pg 1 for expression of curl in cylindrical co-ordinates]

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{A}) = -\frac{\partial A_z}{\partial s}\hat{\varphi} = -\frac{\partial}{\partial s}\left(\frac{k}{s}\right)\hat{\varphi} = \frac{k}{s^2}\hat{\varphi}$$

$$\therefore \boxed{\vec{J} = \frac{1}{\mu_0}\frac{k}{s^2}\hat{\varphi}}$$

3.



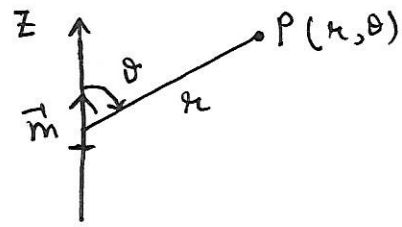
(a) Magnetic dipole moment is given by the strength of current times the area enclosed by the current loop

$$\vec{m} = I(\pi R^2)\hat{z}$$

[direction of $\vec{m}$ is given by right hand rule]

(b) The <u>approximate</u> magnetic field at points far from the origin is given by the dipole potential - as because, from multipole expansion of vector potential, monopole contribution vanishes - the first <u>non-zero</u> contribution to the vector potential is dipole potential $\vec{A}_{dip} = \frac{\mu_0}{4\pi}\left(\frac{\vec{m}\times\hat{r}}{r^2}\right)$ —(i) Now if $\vec{m}$ aligned along z axis (special case), the magnetic field (i.e $\vec{\nabla}\times\vec{A}_{dip}$) or dipole field at a point (r,θ) is given by

$$\vec{B}_{dip}(r,\theta) = \frac{\mu_0}{4\pi}\frac{1}{r^3}\left[2m\cos\theta\,(\hat{r}) + m\sin\theta\,(\hat{\theta})\right] \quad \text{—— (ii)}$$



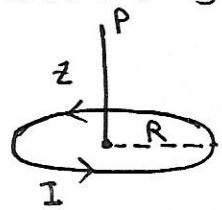
This is discussed in class and can be easily obtained by taking $\vec{\nabla}\times\vec{A}_{dip}$ in spherical polar co-ordinates [see Griffiths, p54]

So approximate magnetic field at points far from origin is eqn (ii) - its approximate because total $\vec{B}$ can contain contribution from quadrupoles, octopoles, ... etc which becomes dominant as r decreases but at ~~distances~~ larger distances, dipole field is the most dominant contribution.

(C) Clearly at points on Z - axis, $\theta = 0$, $\hat{r} = \hat{z}$ and $r = z$. Putting these in (ii)

$$\vec{B}_{dip}(r,\theta) \simeq \vec{B}(r,\theta) = \frac{\mu_0}{4\pi}\frac{1}{z^3}\,2m = \frac{\mu_0}{4\pi}\frac{1}{z^3}\,2(I\times\pi R^2)$$

$$= \frac{\mu_0}{4\pi}\left(\frac{2\pi I R^2}{z^3}\right)$$ which is the same expression as obtained for $\vec{B}$ due to current in circular loop (discussed in class), where



$$B(z) = \frac{\mu_0}{4\pi}\frac{2\pi I R^2}{(R^2+z^2)^{3/2}} = \frac{\mu_0}{4\pi}\left(\frac{2\pi I R^2}{z^3}\right) \quad (z \gg R)$$

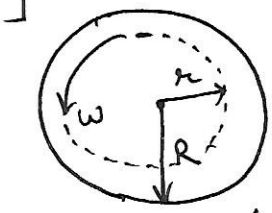
4. $m = I \times \pi r^2$ [for a ring of radius 'r']

Roughly speaking 'I' can be regarded as

$\sigma v\, dr$ $\quad [I = \lambda v, \quad q \sim \lambda dl \sim \sigma da$
$\lambda \sim \sigma dr]$

$dI =$

$\sigma r \omega\, dr \quad [\because v = r\omega]$

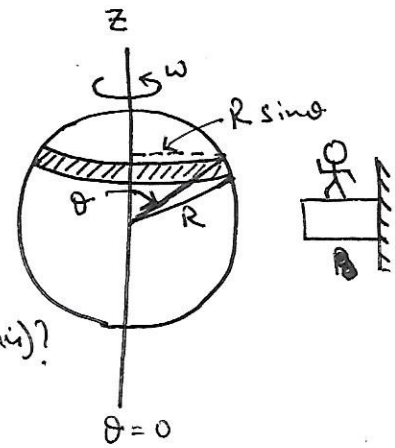


phonograph (uniformly charged disk with surface charge, σ)

$$m = \int dm = \int_0^R \sigma \omega r\, dr \times \pi r^2 = \pi \sigma \omega \int_0^R r^3 dr$$

$$\boxed{m = \pi \sigma \omega \frac{R^4}{4}}$$

5. To find the magnetic moment of the uniformly charged rotating sphere, consider an elementary ring of $R\sin\theta$ radius and width $= R\,d\theta$. What is the equivalent current if the angular velocity is ω about the line $\theta = 0$ (z-axis)?



We may consider an observer who is stationary in the laboratory observing the shaded ring as it rotates. The total charge on the ring is $2\pi R\sin\theta \times R\,d\theta$ and it rotates past the observer in $\frac{\omega}{2\pi}$ times in unit time. Thus the current is

$$I = \frac{dq}{dt} = \frac{2\pi R^2 \sin\theta\, d\theta}{\frac{2\pi}{\omega}} = \omega R^2 \sin\theta\, d\theta$$

Since area of ring is $\pi(R\sin\theta)^2$, dipole moment of shaded ring is

$$m = \int_{\theta=0}^{\pi} \omega R^2 \sin\theta\, d\theta\, \pi R^2 \sin^2\theta = \pi \sigma \omega R^4 \int_0^{\pi} \sin^3\theta\, d\theta$$

$$\boxed{\vec{m} = \frac{4}{3} \pi \sigma \omega R^4\, \hat{z}}$$

If you calculate dipole potential, $\vec{A}_{dip} = \frac{\mu_0}{4\pi}\left(\frac{\vec{m}\times\hat{r}}{r^2}\right)$

$$\vec{m} = \frac{4}{3}\pi\sigma\omega R^4\,\hat{z} = \frac{4}{3}\pi\sigma\omega R^4\left(\cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}\right)$$

$$\left[\hat{z} = \cos\theta(\hat{r}) - \sin\theta(\hat{\theta})\right]$$

$$\vec{A}_{dip} = \frac{\mu_0}{4\pi}\frac{1}{r^2}\frac{4}{3}\pi\sigma\omega R^4 \sin\theta\,\hat{\varphi} \quad \left[\hat{\theta}\times\hat{r} = -\hat{\varphi}\right]$$

$$\boxed{\vec{A}_{dip} = \frac{\mu_0\sigma\omega R^4}{3}\left(\frac{\sin\theta}{r^2}\right)\hat{\varphi}} \quad \text{——— (iii)}$$

Now, an exact expression of $\vec{A}$ same as above due to spinning charged sphere would be obtained by direct integration

$$\vec{A} = \frac{\mu_0}{4\pi}\int\frac{\vec{K}\,da'}{r} \quad \text{[see Griffiths, Ex 5.11 p245]}$$

which proves that the expression of $\vec{A}_{dip}$ obtained in (iii) above is exact.